What Gets Measured: A Study of Productivity and Mental Health Metrics for Software Development Teams

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***Abstract*—Software development relies on productivity metrics, while the industry faces concerns about mental health and burnout. This study investigates the association between productivity measurement practices and mental health discourse, hypothesizing that individual-tracking metrics are associated with more negative discourse than systemic, team-level metrics. We use a mixed-methods design combining the analysis of 278,131 Reddit messages from five developer communities (2021–2024) using VADER sentiment analysis with a practitioner survey ($N = 24$). The mining results show that productivity discourse is dominated by evaluation practices and individual-output metrics. Systemic metrics are associated with 21.9% negative discourse, compared with 30.8% for individual metrics. A logistic regression further shows that mentioning a systemic metric is associated with 24% lower odds of negative discourse (OR = 0.76, $p = 0.003$). Preliminary survey results also indicate a perceived relationship between productivity and mental health. Overall, the findings suggest that how productivity is measured is associated with the tone of developers' discourse about work and well-being.**

***Index Terms*—developer productivity, mental health, software metrics, burnout, sentiment analysis, empirical software engineering.**

I. Introduction

The software industry has long sought to quantify the work of building software. From Lines of Code (LOC) and commit counts to Agile velocity points and individual Key Performance Indicators (KPIs), organizations have adopted a succession of metrics intended to make development visible and manageable. Yet measurement in software engineering is fragile [1]. A metric that is easy to track is rarely the same as a metric that reflects real value.

In parallel, the technology sector has confronted a sustained mental health crisis. Burnout, recognized by the World Health Organization as an occupational phenomenon resulting from chronic, unmanaged workplace stress [2], is widespread among software practitioners [3], and the emotional state of developers has measurable consequences for their work [4], [5]. This crisis was sharpened by the post-pandemic restructuring of the industry: waves of layoffs beginning in late 2022, and Return-to-Office (RTO) mandates that removed flexibility many developers had come to rely on. Despite growing awareness, organizations typically treat the two domains in isolation. Productivity is measured to evaluate efficiency, while well-being is delegated to optional human resources surveys, with little attempt to relate one to the other.

This study argues that treating the two domains separately is a mistake. Our central hypothesis is that metrics oriented toward *individual* tracking and surveillance, such as individual velocity, individual OKRs, and lines of code, are associated with more negative discourse among the developers subject to them, whereas *systemic* metrics that measure the delivery pipeline, such as the DORA metrics [6], are perceived as less personally directed and coexist with healthier discourse. The design is observational, so we test this as an *association* between a metric's framing and the surrounding discourse, not as a causal claim that the metrics produce the distress. Concretely, we ask which metrics the industry uses and how their *framing* relates to the well-being of the people measured.

A. Objectives and Research Questions

The objective of this study is to catalog the productivity and mental health metrics that software professionals use or consider desirable, and to characterize the gap between what is currently measured and what practitioners believe *should* be measured. We pursue this through five research questions, answered by two complementary sources of evidence: a large-scale mining of developer discourse on Reddit, which observes what developers *discuss* unprompted (RQ1–RQ3), and a struc-

tured practitioner survey, which asks them directly what they *use* and value (RQ4–RQ5).

RQ1 *How frequently are productivity and mental health metrics discussed by developers in practice?* (Reddit)

RQ2 *What sentiments do developers express toward different metrics?* (Reddit)

RQ3 *How have mental health and productivity metrics been discussed over time?* (Reddit)

RQ4 *Which productivity and mental health metrics are currently used by developers, and how do they perceive their effectiveness?* (Survey)

RQ5 *Which productivity and mental health metrics do developers consider desirable but underused or missing?* (Survey)

B. Approach and Contributions

To answer these questions we adopt a mixed-methods design that draws on two independent sources of evidence. The primary source is a large-scale mining of developer discourse: we analyze 278,131 messages from five developer communities on Reddit (2021–2024), drawn from public Reddit archives, and apply lexicon-based sentiment analysis to quantify how the community discusses different metrics. This captures unsolicited discourse, free of the social desirability bias common to workplace surveys. Complementing it, a structured practitioner survey elicits, in a controlled form, the metrics developers actually use, the effectiveness they ascribe to them, and the metrics they wish existed. The two sources were designed independently and address complementary questions, allowing the survey to corroborate, on the central hypothesis, what the observational findings suggest.

This study makes two contributions: (i) an empirically grounded catalog of productivity and mental health metrics as perceived by practitioners, derived from spontaneous developer discourse and a complementary survey instrument; and (ii) quantitative evidence that the framing of a metric, individual versus systemic, is associated with the emotional valence of the discussion surrounding it, with systemic metrics such as DORA attracting clearly less negative discourse than individual-tracking metrics.

II. Background and Related Work

A. The Evolution of Software Productivity Metrics

Efforts to quantify software work have historically relied on measures inherited from industrial production, with code and development activity treated as observable proxies for productivity. Lines of Code and commit counts became particularly common because of their simplicity and availability. However, their relationship with productive value is weak: when measures become targets, developers can optimize for the metric rather than for the underlying objective, illustrating the mechanism described by Goodhart's Law [1]. DeMarco and Lister similarly argued that productivity in software development is shaped by organizational, environmental, and human factors that cannot be reduced to output volume [7].

The limitations of output-based measures also extend to Agile practices. Story points and velocity are commonly used to support estimation, planning, and forecasting within Agile teams, but their use as performance measures introduces important methodological challenges [8]. Their organizational use, however, has frequently transformed them into performance indicators, encouraging teams to increase reported velocity independently of delivered value [8]. This distinction is relevant because a measure designed to support coordination can acquire a different function when incorporated into systems of evaluation and accountability.

More recent approaches have consequently shifted attention from individual output to properties of the development system. DORA proposes four delivery-performance measures—Deployment Frequency, Lead Time for Changes, Mean Time to Recovery, and Change Failure Rate—intended to characterize software delivery performance at the system level [6]. The SPACE framework further conceptualizes developer productivity as multidimensional, encompassing Satisfaction and Well-being, Performance, Activity, Communication and Collaboration, and Efficiency and Flow [9]. The DevEx framework similarly emphasizes developers' day-to-day experience as an important determinant of software delivery outcomes [10]. Together, these approaches challenge the assumption that productivity can be adequately represented by a single measure of individual output.

Empirical studies reinforce this multidimensional perspective. An industry study at Google found no single metric capable of adequately representing developer productivity and cautioned against interpreting individual measures in isolation [11]. Murphy-Hill et al. further showed that factors such as autonomy, enthusiasm, and peer support are associated with developers' perceptions of productivity, highlighting the importance of contextual and social conditions [12]. Similarly, Meyer et al. found that developers conceptualize productivity in personal and context-dependent terms that are poorly captured by simple activity counts [13]. Thus, the evolution from LOC and commits toward systemic and multidimensional frameworks reflects a broader reconsideration of what constitutes productivity in software engineering.

B. Mental Health and Burnout in the Technology Sector

Software productivity cannot be considered independently of developers' well-being. Recent empirical studies in software engineering have increasingly examined the relationship between mental health, emotional states, and work outcomes, suggesting that developers' psychological well-being is intertwined with how they experience and perform their work [14]–[16]. This relationship is particularly relevant in the context of burnout. The World Health Organization defines burnout as an occupational phenomenon resulting from chronic workplace stress that has not been successfully managed, characterized by exhaustion, cynicism, and reduced professional efficacy [2]. These dimensions are consistent with the foundational model proposed by Maslach et al. [3].

Empirical research in software engineering provides evidence of an association between developers' emotional states and their work outcomes. Graziotin et al. show that negative affect is associated with cognitive and behavioral consequences relevant to software development, including reduced focus and lower perceived productivity [4], [5]. At the level of software artifacts, Ortu et al. report an association between negative emotions expressed in issue discussions and longer issue-fixing times [17]. M"antyl"a et al. further demonstrate that emotional signals extracted from developers' written communication can provide indicators of burnout and productivity loss [18]. Storey et al. extend this line of work by proposing a relationship in which job satisfaction and perceived productivity may reinforce one another [19]. Collectively, these studies suggest that productivity and well-being are not independent dimensions of software development.

Written developer communication therefore provides a potentially relevant source for studying this relationship. At the same time, sentiment analysis in software engineering presents methodological challenges. Jongeling et al. demonstrate substantial disagreement among sentiment-analysis tools and between automated classifications and human judgments in software engineering communication [20]. Similar limitations have been reported across multiple software engineering datasets [21], while Novielli et al. show that adapting sentiment models to software engineering data can improve performance but does not eliminate tool dependency [22]. These findings indicate that sentiment results should not be interpreted independently of the analytical method used. Accordingly, the present study compares lexicon-based sentiment scores with predictions from a transformer-based model rather than relying on a single sentiment-analysis approach.

Beyond individual emotional states, organizational conditions also influence developers' well-being. Edmondson's concept of psychological safety describes a shared perception that a team is safe for interpersonal risk-taking and has been associated with learning and performance in organizational settings [23]. This perspective is particularly relevant to productivity measurement because systems that use individual metrics for comparison, ranking, or evaluation may alter the social conditions under which developers work. In contrast, the SPACE framework explicitly treats satisfaction and well-being as dimensions of productivity rather than as competing outcomes [9]. The way productivity is measured may therefore constitute part of the organizational environment through which productivity and well-being interact.

C. The Gap: Measuring Work Without Measuring the Worker

The literature on software productivity and the literature on developer well-being have developed substantially, yet their intersection remains comparatively underexplored. Productivity research has increasingly recognized the limitations of individual activity measures and incorporated contextual, social, and experiential dimensions [9], [10]. Conversely, research on developer well-being has established associations between emotional states, working conditions, burnout, and software development outcomes [4], [18], [19]. What remains less understood is how developers experience productivity measurement itself, particularly when metrics become mechanisms of individual evaluation, surveillance, or performance comparison.

Existing approaches also face methodological limitations. Survey-based studies provide direct access to developers' perceptions but may be affected by social desirability and retrospective bias. Conversely, organizational productivity metrics provide large-scale behavioral evidence but generally lack information about developers' emotional and experiential states. Unsolicited developer discourse offers a complementary source of evidence because it captures naturally occurring accounts of work experiences without requiring participants to frame their experiences in response to a research instrument.

This study addresses this gap by combining large-scale analysis of developer discourse with a targeted survey. The discourse analysis examines how productivity measurement, performance expectations, and well-being are discussed in naturally occurring software engineering communication, while the survey provides direct evidence of developers' perceptions of these relationships. Together, these sources enable an empirical examination of the relationship between productivity measurement and developer well-being, complementing existing frameworks that conceptualize productivity as a multidimensional and systemic phenomenon.

III. Study Design

This study investigates which metrics for measuring productivity and mental health are used or considered desirable by software development professionals. We focus on *perceived* utility and idealized alternatives rather than metrics measured against objective performance data, drawing on two independently designed sources: (1) a large-scale mining of developer discourse on Reddit, which observes what practitioners say unprompted, and (2) a structured practitioner survey, which asks them directly. The two address complementary questions that converge on the study's central hypothesis. Figure 1 summarizes the five-step design.

A. Research Questions

We organize the study around five research questions (Section I-A), answered by the two independent sources. The Reddit corpus observes how metrics are actually *discussed*, answering RQ1 (how frequently each metric is discussed, from the volume split and the catalog of Sections V and VIII), RQ2 (the sentiment each metric attracts, from the framing and community analyses of Sections VI and VII), and RQ3 (how that discussion evolves over 2021–2024, Section VIII). The survey asks developers directly which metrics they *use* and value, answering RQ4 (the metrics currently adopted and their perceived effectiveness) and RQ5 (the metrics considered desirable but underused or missing). The observational evidence (RQ1–RQ3) and the survey questions (RQ4–RQ5) are thus complementary: the corpus shows what developers

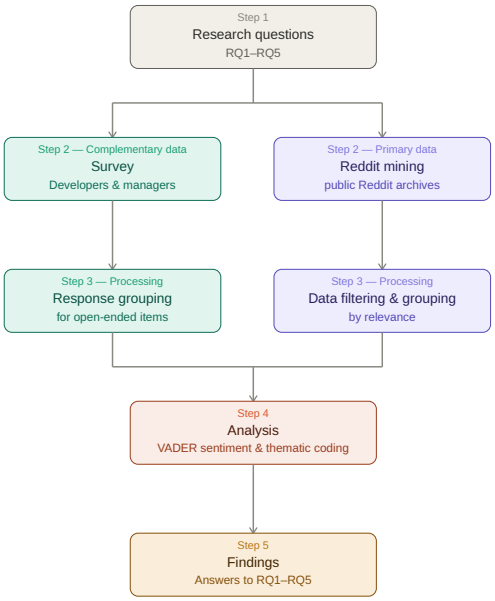


Fig. 1. Study design workflow.

say unprompted, and the survey corroborates it with solicited responses. The five questions are stated in full in Section I-A.

B. Reddit Mining

a) Rationale.: Reddit is organized into topic-focused communities (subreddits), which makes it well suited to targeting software professionals as they discuss a specific subject. It is also one of the principal forums where technology professionals discuss their careers, working conditions, and tools anonymously, free of the professional self-censorship typical of platforms such as LinkedIn. This anonymity reduces the social desirability bias that affects direct workplace surveys, exposing the relationship between metrics and well-being as it is spontaneously expressed. Mining developer discourse to study emotion is an established empirical method [17], [18]. We analyze both posts and their comments: where a post often only raises a topic, the comments carry the community's reaction to it, and including them substantially enlarges the sample of opinionated text about each metric.

b) Corpus.: Data were obtained from the complete public Reddit archives (the Pushshift dumps [24], distributed as per-subreddit `zstandard` files via Academic Torrents), which provide every post and comment of each community. We targeted five subreddits frequented by software professionals across different roles, seniority levels, and backgrounds (Table I). The collection window spanned 1 January 2021 to 31 December 2024, deliberately encompassing the post-pandemic layoffs and return-to-office (RTO) mandates. After filtering, the corpus comprised 25,120 posts and 253,011 comments (278,131 messages).

c) Query filters.: Messages were retrieved using two keyword query groups corresponding to the research domains

TABLE I
THE FIVE TARGET SUBREDDITS, THEIR FOCUS, AND THEIR POST COUNTS IN THE FILTERED CORPUS (2021–2024).

Subreddit	Focus / role	Posts
`r/cscareerquestions`	General software careers, hiring, and early-career questions	16,446
`r/ExperiencedDevs`	Senior practitioners; team, process, and management topics	4,178
`r/devops`	Infrastructure, operations, reliability, and tooling	2,518
`r/softwareengineering`	Engineering practice, design, and methodology	1,070
`r/womenintech`	Experiences of women and underrepresented groups in technology	908
Total		**25,120**

TABLE II
KEYWORD LISTS DEFINING THE TWO QUERY GROUPS (CASE-INSENSITIVE). SHORT OR AMBIGUOUS ACRONYMS (*RTO*, *OKR*, *KPI*, *DORA*, *SLA*, *SLO*, *SLI*, *MTTR*, *Jira*) ARE MATCHED AS WHOLE WORDS.

Mental health	Metrics and productivity
burnout, burned out, exhausted, overwhelmed, anxiety, stress, fear, depression, mental health, therapy, medication, overtime, quitting, micromanagement, imposter syndrome, work-life, toxic, hostile, RTO, return to office	metric(s), measure, measurement, productivity, OKR, KPI, DORA, velocity, story points, code coverage, lines of code, cycle time, throughput, lead time, deployment frequency, change failure, MTTR, SLA, SLO, SLI, four key, flow metric, burndown, kanban, scrum, sprint, Jira, tech debt, technical debt, pull request, performance review, benchmark, tracking

(Table II). The same keyword matching was applied to post text (title and body) and to comment bodies. Each message was assigned to the query group whose terms it matched; when a message matched both groups, the mental-health axis took precedence. This precedence only *disambiguates the axis label* of a doubly matching message; it does not narrow the study to mental health, and the metrics axis still retains every message that did not also raise a mental-health term.

d) Preprocessing.: Filtering was deliberately light, to keep the text close to how it was written. Messages whose body had been deleted or removed by Reddit, or was empty, were dropped (an empty post was kept only if it still carried a positive score, i.e. a title-only thread that nonetheless drew engagement), and only messages inside the 2021–2024 window were retained. Post bodies were truncated to the first 500 characters; titles were kept in full. Before sentiment scoring, the title and body were concatenated and lower-cased. We applied no stop-word removal, stemming, URL or emoji stripping, or de-duplication, as VADER is designed to operate on raw social-media text.

e) Unit of analysis.: Because posts and comments answer different questions, we use them accordingly. Analyses of *volume*, temporal trends, community composition, and the metric catalog use posts, the natural unit of a discussion

thread (the catalog asks what a developer raises in their own post, not what accumulates in a reply chain). Analyses of *sentiment* use the full corpus (posts and comments): comments are where reaction concentrates, and the pooled sample gives the statistical power that per-metric comparisons require. We report posts and comments separately as a robustness check (Table V).

f) Sentiment analysis.: To quantify community perception we applied VADER, a rule-based sentiment model validated for social-media text [25]. Each message received a compound polarity score in $[-1, 1]$; following the standard threshold, messages scoring below -0.05 were classified as negative. The framing categories of Section VI and the per-metric sentiment comparisons are formed by matching a curated list of metric names and synonyms against the lower-cased text. For short or ambiguous terms (e.g., *SLA*, *OKR*, *LOC*) we required a word-boundary match, so that an acronym matches the whole word and not a substring of an unrelated term (for instance, *LOC* inside "block" or *SLA* inside "Slack"), reducing false positives; longer, unambiguous phrases (e.g., *technical debt*, *performance review*) are matched as substrings.

Off-the-shelf sentiment tools are known to disagree when applied to software engineering text [20]–[22], so we do not rest the sentiment claims on a single model. Every framing-matched message was also scored by a second, independent model: a RoBERTa transformer fine-tuned for sentiment classification on social-media text [26], run locally on the same text VADER received. The transformer emits a negative/neutral/positive label; to compare the two, VADER compound scores are mapped to the same three classes with the standard ± 0.05 thresholds. We report agreement between the models as raw percent agreement and as Cohen's kappa [27], and re-estimate the central framing comparison with the transformer's labels as the outcome (Section VI-A).

g) Statistical analysis.: Reddit messages are not independent observations: comments nest inside the thread of their parent post and share its topic and tone, so tests that treat each message as independent overstate the effective sample size. Our primary inference is therefore a logistic regression of the binary negativity indicator on the framing category, with the subreddit, the year, the message type (post or comment), and the log of the text length as controls, and with standard errors clustered by thread [28]. The model is fitted on the messages that match exactly one framing category (97% of the framing-matched documents), so framing enters as a three-level factor with person/output as the reference level; a robustness fit on the full, non-exclusive set with indicator variables is reported alongside it. As a complement that needs no clustering correction, we also run the central system-versus-individual contrast on posts alone, since a post is the top of its own thread. The unadjusted chi-square and Kruskal–Wallis tests are retained as descriptive statistics.

h) Cataloging metrics and practices.: The catalog of Section V is built by open extraction, not from a fixed keyword list. Each post is passed through a locally-run large language model (Gemma 3, 4B parameters, at temperature 0 for reproducibility). We use a 4B model because it runs locally on a single consumer GPU, which keeps the analysis cheap to reproduce and lets the posts be processed on-device rather than sent to an external API. The model is prompted to return, as structured JSON, only the productivity metrics, frameworks, management practices, and well-being indicators that the text explicitly names. The raw extractions are then consolidated (surface forms canonicalized against a seed taxonomy with fuzzy matching, obvious noise removed, and distinct-post counts aggregated per item), after which a manual curation pass retains the study-relevant items and labels each as pertaining to *productivity*, *mental health*, or *both*. To validate it, we run a fixed-keyword regex catalog over the same posts as a baseline: a hand-compiled list of 22 canonical metrics and practices (story points, velocity, DORA, OKR, KPI, performance review, lines of code, and similar named items drawn from the query vocabulary of Table II). The model recovers all 22 (100% coverage) and additionally surfaces items that no a-priori list anticipated, such as *imposter syndrome*, *toxic environment*, and *return-to-office*. This produced the curated catalog reported in Chapter IV.

TABLE III
DEFINITION OF THE THREE METRIC-FRAMING CATEGORIES USED IN THE SENTIMENT ANALYSIS. A MESSAGE BELONGS TO A CATEGORY IF ITS LOWER-CASED TEXT MATCHES ANY OF THE CATEGORY'S TERMS (ACRONYMS MATCHED AS WHOLE WORDS). THE CATEGORIES ARE NOT MUTUALLY EXCLUSIVE.

Category	Matching terms
System / pipeline	dora, deployment frequency, change failure, mean time to recover, four key
Person / output	pip, performance improvement plan, performance review, story points, velocity, okr(s), kpi(s), lines of code
Scrum / Agile process	scrum, agile

i) Metric framing categories.: For the sentiment analysis of Section VI, we additionally group messages into three *framing* categories by the metrics they name, contrasting person/output-oriented measurement, system/pipeline-oriented measurement, and Agile process (Table III). We label velocity and story points as person/output-oriented even though, as Section **??** notes, they were designed as team-relative estimation aids: in the practitioner discourse we study they are routinely repurposed as targets applied to the individual (Section IX), which is the framing the categories are meant to capture. Two properties matter when reading the results. First, the categories are *not* mutually exclusive and have no precedence: a message mentioning both *velocity* and *DORA* is counted in both, so the category sizes overlap and do not sum to the corpus (in practice the overlap is small, under 7% of the smaller category in each pair). Second, this categorization is independent of the two query groups of Table II: the query groups decide which messages enter the corpus (with mental-health precedence), whereas the framing categories slice the metric vocabulary *inside* the corpus and carry no precedence.

j) Ethics and anonymization.: In accordance with ethical guidelines for mining user-generated content [29], all author identifiers were replaced with SHA-256 hashes and direct permalinks were removed from the working dataset prior to analysis. This preserves the semantic value of the discussions while protecting the identity of individuals. Only aggregate statistics and non-identifying excerpts of high-visibility public posts are reported.

C. Survey Instrument

The survey corroborates the observational findings, following established guidance for opinion surveys in software engineering [30]. Respondents first pass through an informed-consent gate (the *Termo de Consentimento Livre e Esclarecido*, TCLE), after which the instrument unfolds in four thematic sections (Table IV): *Demographics*, *Productivity*, *Mental Health*, and a closing *Holistic* item. It mixes single- and multiple-choice items, 5-point Likert scales, and open-ended questions. The two multi-select adoption questions (one per domain) are followed, for each option a respondent selects, by a 5-point Likert item rating that specific metric or practice, so perceived effectiveness is captured per item, not in the aggregate. Closed items map the metrics and well-being practices developers currently use and how effective they consider them (RQ4). The open-ended items ask for an unprompted listing of metrics the respondent considers good, the single most useful metric not yet adopted, and the observable signals of compromised mental health; these address RQ5 and feed an inductive thematic analysis [31].

The survey is distributed via snowball sampling through professional networks and developer communities. Participation is voluntary and begins only after the respondent consents through the TCLE; no personally identifying information is required, with the sole exception of an optional e-mail field for respondents who wish to be contacted about the results. We report the responses gathered so far in Section XII; collection remains open, so those results are read as descriptive triangulation of the corpus rather than as a standalone quantitative test.

IV. Results and Discussion of the Research Questions

The Reddit corpus comprises 278,131 messages (25,120 posts and 253,011 comments). Mental health discourse dominates the posts: 16,696 (66%) match the mental health axis against 8,424 (34%) the metrics axis, so well-being is discussed about twice as often as measurement and productivity.

V. What the Community Discusses: A Catalog of Metrics and Practices

This section feeds RQ1 (what gets discussed). Rather than count a fixed keyword list, we catalog what developers *spontaneously name*: each post is passed through a locally-run language model that openly extracts every productivity metric, framework, management practice, or well-being indicator the text explicitly mentions, which we then consolidate and curate into a single inventory (Section III-B). Figure 2 shows the 25 most-named items of the 135 in the curated catalog, each colored by whether it pertains to productivity, to mental health, or to both. Three observations stand out.

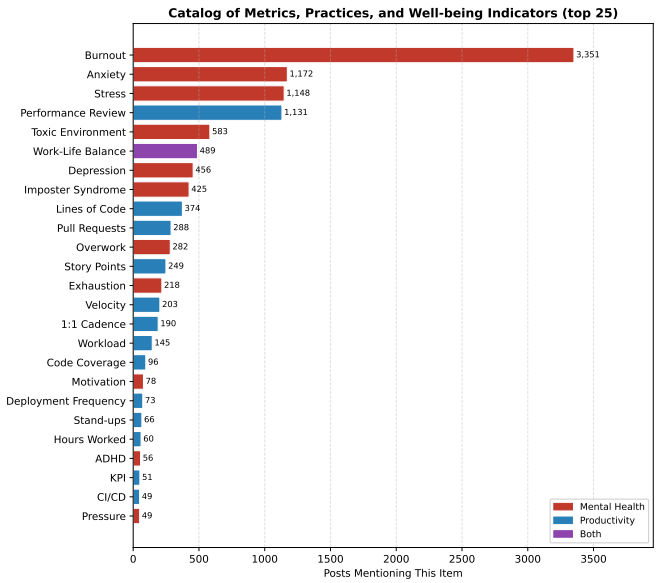


Fig. 2. The metrics, practices, and well-being indicators most often named across the posts (top 25 of 135 curated items), colored by whether each pertains to productivity, mental health, or both. Counts are the number of distinct posts naming the item.

First, well-being terms dominate the head of the catalog. The single most-named item, by a wide margin, is Burnout (3,351 posts), followed by Anxiety (1,172) and Stress (1,148). Across the whole catalog the 29 well-being items account for 7,994 post-mentions against 3,889 for the 88 productivity items (the remaining 18 of the 135 curated items pertain to both domains). Developers thus draw on a narrower but far more frequently invoked mental-health vocabulary, which echoes the roughly two-to-one dominance of well-being seen in the raw post volume.

Second, the productivity items are led by evaluation practices and individual output. The most-named productivity item is a *practice*, Performance Review (1,131 posts), with the recurring manager–developer one-on-one ("1:1 Cadence", 190) also prominent. The classical individual metrics follow: Lines of Code (374), Pull Requests (288), Story Points (249), and Velocity (203). Each of these measures the person or their immediate output; raw output counts as such come further down the list.

Third, the systemic counterpoint to these individual measures is almost absent from the *named* vocabulary. System- and team-level metrics, which gauge the delivery pipeline or the team (for example the DORA delivery set or lead time), are named too rarely to survive curation as catalog items, even though they anchor the framing analysis of Section VI. The metrics developers reach for spontaneously are overwhelmingly individual.

TABLE IV
STRUCTURE OF THE PRACTITIONER SURVEY. THE TWO MULTI-SELECT ITEMS (#8, #11) EXPAND INTO A 5-POINT LIKERT FOLLOW-UP FOR EACH OPTION THE RESPONDENT SELECTS.

#	Question (abridged)	Type	Research goal
Consent			
–	Informed consent to participate (TCLE)	Yes/No	Eligibility gate
Section 1 – Demographics			
1	Main role in software development (11 options)	Single choice	Role moderator
2	Years of experience in that role	Single choice	Seniority moderator
3	Work model (on-site/hybrid/remote)	Single choice	Context for exposure
4	Development team size	Single choice	Team scale
5	Company size	Single choice	Organizational scale
6	Company business model	Single choice	Metric-culture context
Section 2 – Productivity			
7	List up to 3 metrics you consider good for measuring a developer's productivity	Open text	Unprompted ideal (RQ4/RQ5)
8	Which metrics does your team/company use? (9 options)	Multi-select	Adoption (RQ4)
8a	*Per selected metric:* does it represent the developer's real productivity?	Likert 1–5	Perceived effectiveness (RQ4)
9	Most useful metric not yet widely adopted?	Open text	Desired/missing (RQ5)
Section 3 – Mental Health			
10	Does your company track developer well-being?	Single choice	Well-being tracking (RQ4)
11	Which practices/tools are used? (8 options)	Multi-select	Adoption (RQ4)
11a	*Per selected practice:* is it effective at identifying/preventing mental-health problems?	Likert 1–5	Perceived effectiveness (RQ4)
12	Signals that a developer's mental health is compromised by work	Open text	Practitioner signals (RQ5)
Section 4 – Holistic			
13	How related are productivity and mental health day-to-day? (independent $\rightarrow$ inseparable)	Likert 1–5	Tests core hypothesis (closing item)

VI. SENTIMENT BY METRIC FRAMING: INDIVIDUAL, SYSTEMIC, AND PROCESS

This section addresses RQ2. The central hypothesis predicts that individually-framed metrics attract more negative discourse than systemic ones, with Scrum/Agile process as a third point of contrast. We grouped messages into the three framing categories defined in Table III and computed, for each, the mean sentiment (the average VADER compound score, from -1 to $+1$) and the negativity rate (the share of messages scoring below -0.05), as shown in Table V.

The result supports the hypothesis. System/pipeline metrics, which gauge the delivery pipeline or the team instead of the individual (the DORA set being the clearest example), form the least negative category (21.9% negativity) and the most positive in mean sentiment ($+0.289$). Person/output metrics and Scrum/Agile process are both close to 31% negativity, roughly nine percentage points worse. The negativity rate is the primary signal of the contrast: because most messages mix positive and negative language, the means compress into a narrow positive band ($+0.18$ to $+0.29$), while the share of outright negative messages separates the categories far more sharply. The ordering is identical when posts and comments are examined separately. Comments are more negative throughout, as expected, since reaction concentrates there, but the gap between the system/pipeline and person/output framings persists in both.

We tested the association formally. The unadjusted tests agree with the table: a chi-square test of independence over the three categories gives $\chi^2 = 29.7$ ($\mathrm{df} = 2$, $p < 0.001$), the central system/pipeline-versus-person/output contrast on its own gives $\chi^2 = 28.4$ ($\mathrm{df} = 1$, $p < 0.001$), and a Kruskal–Wallis test on the underlying sentiment distributions gives $H = 68.0$ ($p < 0.001$). These tests, however, treat every message as an independent observation, when in fact comments belong to the thread of the post they respond to, and they say nothing about confounds such as which community or year a message comes from. We therefore strengthen the analysis in two ways. First, restricting the system-versus-individual contrast to posts alone, where each message heads its own thread and no nesting exists, preserves the association ($\chi^2 = 5.9$, $\mathrm{df} = 1$, $p = 0.015$, $N = 1{,}271$, $\phi = 0.07$). Second, and as the primary test, we fit a logistic regression of the negativity indicator on the framing category, controlling for subreddit, year, message type, and log text length, with standard errors clustered by thread [28]. The model is fitted on the 32,682 messages that match exactly one framing (971 multi-framing messages are excluded), spread over 12,761 threads (Table VI).

Comparing messages from the same subreddit, the same year, the same type, and the same length, a message naming a system/pipeline metric still has about 24% lower odds of being negative than one naming a person/output metric (OR $= 0.76$, 95% CI $[0.63, 0.91]$, $p = 0.003$), while the Scrum/Agile process category does not differ from person/output (OR $= 1.04$, $p = 0.24$). Refitting on the full, non-exclusive framing set with indicator variables barely moves the estimate (OR $= 0.74$, $p < 0.001$). With tens of thousands of messages even a trivial difference reaches significance, so the effect size still

TABLE V

SENTIMENT BY METRIC FRAMING, WITH THE POSTS-ONLY AND COMMENTS-ONLY BREAKDOWN. CATEGORIES ARE DEFINED IN TABLE III AND ARE NOT MUTUALLY EXCLUSIVE. SENT. IS THE MEAN VADER COMPOUND SCORE; NEG. IS THE NEGATIVITY RATE.

	Full corpus			Posts			Comments		
Category	**N**	**Sent.**	**Neg.**	**N**	**Sent.**	**Neg.**	**N**	**Sent.**	**Neg.**
System / pipeline metrics	811	+0.289	21.9%	95	+0.310	14.7%	716	+0.286	22.9%
Person / output metrics	16,128	+0.208	30.8%	1,176	+0.271	26.6%	14,952	+0.203	31.2%
Scrum / Agile process	17,688	+0.176	30.9%	1,012	+0.230	27.6%	16,676	+0.173	31.1%

TABLE VI

LOGISTIC REGRESSION OF THE NEGATIVITY INDICATOR (VADER COMPOUND < -0.05) ON THE METRIC FRAMING, CONTROLLING FOR SUBREDDIT, YEAR, MESSAGE TYPE, AND LOG TEXT LENGTH. STANDARD ERRORS ARE CLUSTERED BY THREAD; REFERENCE LEVEL IS PERSON/OUTPUT. $N = 32{,}682$ MESSAGES IN 12,761 THREADS.

Framing (vs. person/output)	**Odds ratio**	**95% CI**	p
System / pipeline	0.76	[0.63, 0.91]	0.003
Scrum / Agile process	1.04	[0.98, 1.10]	0.240

matters: Cramér's V for the three-category table is only 0.03 ($\phi = 0.04$ for the two-category contrast), a small association. What carries the finding is the magnitude and stability of the gap itself: a roughly nine-point difference in negativity (21.9% against 30.8%) that survives the covariate adjustment, the thread-level clustering, and the posts-only restriction. The design is observational, so we read this as an association, not as evidence that a metric's framing causes the surrounding sentiment.

Where Table V groups messages by category, Table VII breaks the same corpus down by individual metric, for a thematically chosen set that spans the person/output–system/pipeline contrast. Five of the six belong to the *person/output* framing (Lines of Code, Velocity/Story Points, KPIs, OKRs, and Commits/PRs); *Throughput / DORA* is the *system/pipeline* one. The matching here is per metric and looser than the category definition of Table III: the *Throughput / DORA* row counts any message naming *throughput* or *dora*, so its 1,504 messages exceed the 811 of the stricter System/pipeline category, which keys on the full DORA delivery family. The per-metric view sharpens the category result. *Velocity / Story Points* is the most negatively discussed metric (31.1% negativity), consistent with its role as the Agile estimate most often repurposed into an individual target, and Lines of Code follows at 28.3%, while the *Throughput / DORA* cluster is again the least negative (20.9%) and most positive (+0.315). The small *Commits / PRs* set (136 messages) is included for thematic completeness, not for its weight.

A. Robustness to the Choice of Sentiment Model

VADER is a lexicon-based model, and in software engineering text the choice of sentiment tool alone can change a study's conclusions [20]. To check that the framing result is not an artifact of one tool, we re-scored every framing-matched message with the RoBERTa transformer described in Chapter III [26]. On the binary negative-versus-not decision that the negativity rate rests on, the two models agree on 69.5% of the 33,653 framing-matched messages (Cohen's kappa $= 0.35$); over the three classes (negative, neutral, positive) agreement drops to 37.9% ($\kappa = 0.18$), driven mainly by the transformer assigning a negative label to many messages that VADER places in its wide neutral band.

The disagreement, however, does not affect the finding (Table VIII). Under RoBERTa the framing gap widens: system/pipeline messages are negative in 19.0% of cases against 43.7% for person/output, a gap of about 25 points, compared with about 8 under VADER. Refitting the regression of Table VI with the RoBERTa label as the outcome gives OR $= 0.36$ (95% CI $[0.28, 0.47]$, $p < 0.001$) for system/pipeline versus person/output. The VADER-based estimate is therefore the conservative one, and the direction of the association does not depend on the choice of sentiment model.

VII. DISCOURSE DIFFERS SHARPLY ACROSS COMMUNITIES

The framing result of Section VI holds over the whole corpus, but the five subreddits are distinct communities with different focuses (Table I), and it is worth asking whether sentiment toward each axis varies by where it is discussed. Figure 3 maps this with two panels, one per query axis. In each panel a bubble is a subreddit; its horizontal position is the share of that subreddit's posts on the axis, its height is the mean VADER sentiment of *those* posts, its size is their number, and its color is their negativity rate. Conditioning sentiment on the post type, rather than on the subreddit as a whole, lets the two axes be read independently.

The mental-health panel (left) orders the communities along a single axis: from `r/devops`, where only 26% of posts concern well-being, to `r/cscareerquestions` (77%) and `r/womenintech` (87%). Along it, the sentiment of mental-health posts falls ($+0.26 \rightarrow +0.11$) and their negativity rises (30% $\rightarrow$ 42%): the more a community is *about* well-being, the more negative that well-being talk becomes. The productivity panel (right) is not a mirror image. Although a subreddit's metrics share is roughly the complement of its mental-health share, the sentiment of its *metrics* posts is a separate quantity: `r/devops` is by far the most positive about metrics (+0.39, 16% negativity), yet `r/cscareerquestions` discusses metrics far more positively (+0.27) than it discusses well-

TABLE VII

SENTIMENT FOR A THEMATICALLY SELECTED SET OF METRICS SPANNING THE PERSON/OUTPUT–SYSTEM/PIPELINE CONTRAST (FULL CORPUS), ORDERED BY MENTION VOLUME. COLUMNS MATCH TABLE V.

Metric	Framing	N	Sent.	Neg.
Velocity / Story Points	Person/output	5,617	+0.197	31.1%
Lines of Code (LOC)	Person/output	3,550	+0.219	28.3%
Throughput / DORA	System/pipeline	1,504	+0.315	20.9%
KPIs	Person/output	897	+0.276	26.2%
OKRs	Person/output	333	+0.282	26.7%
Commits / PRs	Person/output	136	+0.287	22.8%

TABLE VIII

NEGATIVITY RATE BY FRAMING UNDER EACH SENTIMENT MODEL, OVER THE MESSAGES MATCHING EXACTLY ONE FRAMING CATEGORY. THE SYSTEM-VERSUS-INDIVIDUAL CONTRAST HOLDS UNDER BOTH MODELS AND IS LARGER UNDER ROBERTA.

Category	N	Neg. (VADER)	Neg. (RoBERTa)
System / pipeline	742	22.1%	19.0%
Person / output	15,174	30.5%	43.7%
Scrum / Agile process	16,766	30.6%	41.9%

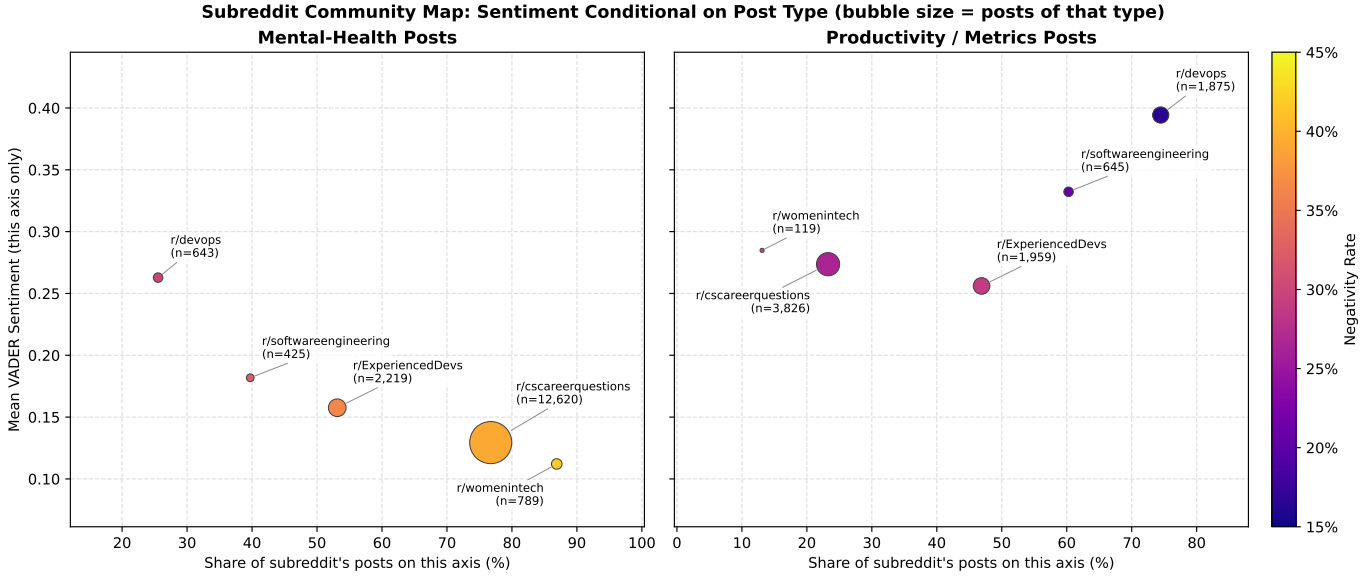


Fig. 3. Community map of mental-health posts (left) and productivity/metrics posts (right) per subreddit; the bubble encoding is described in the text above.

being (+0.13), so the two panels carry genuinely different information.

This reflects a difference in what each community is *for*, not a claim about the well-being of its members. Communities oriented toward systemic technical work surface fewer mental-health topics and a more positive tone; communities oriented toward careers and workplace experience host more discussion of stress and evaluation. The spaces closest to systemic metrics are also where measurement is discussed with the least negativity, but role, demographics, and topic are confounded, so no causal reading is warranted, and we make no claim about any specific metric within any specific community.

VIII. TEMPORAL TRENDS

Over 2021–2024 both discourse axes grew, but at very different rates. Mental health posts grew by 22.5% (4,082 → 5,001) while metrics-and-productivity posts grew by 68.5% (1,637 → 2,759). The conversation about measurement is accelerating about three times as fast as the conversation about its human cost (Figure 4). The monthly series also bears the signature of industry events: a rise in volume from late 2022 (the onset of mass layoffs) and a sharp peak in RTO discussion in 2023.

IX. SURVEILLANCE AND PIPS: WHAT THE TAXONOMIES OMIT

Two recurring patterns fall outside conventional metric taxonomies yet are central to how developers experience measurement.

a) Employee surveillance.: 1,119 posts discuss employer tracking or monitoring of developer activity (screen capture, activity logging, keystroke monitoring) deployed as a productivity proxy. The single highest-scoring post in the entire metrics axis is an account of discovering such tracking.

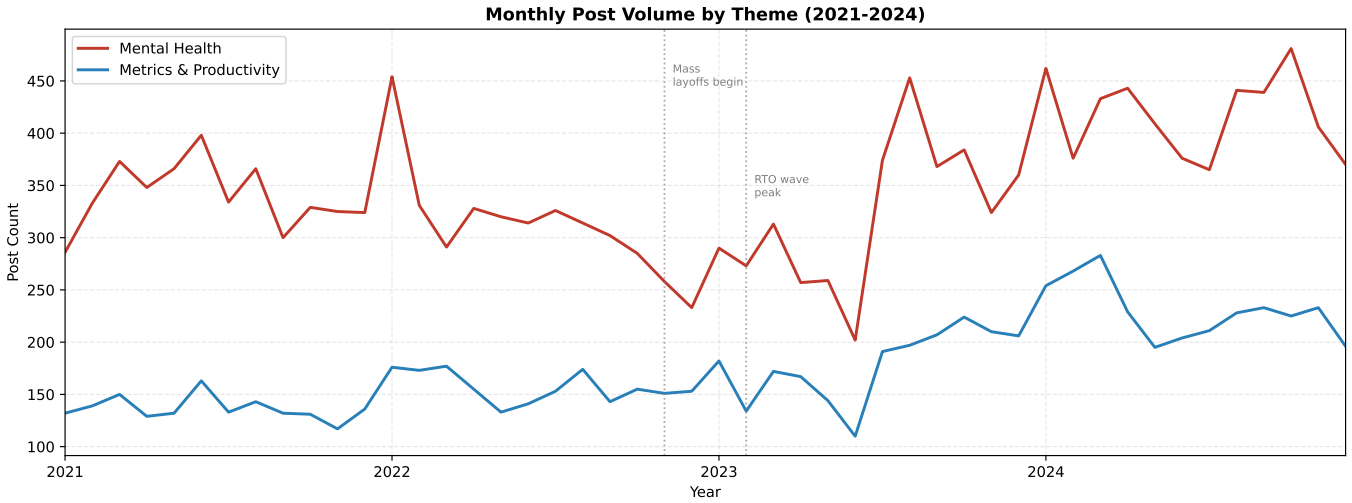


Fig. 4. Monthly post volume by theme, 2021–2024, with industry-event markers.

Surveillance is the limiting case of the individual metric: measurement of the person taken to its most intrusive extreme, and it provokes correspondingly strong reactions.

b) Performance Improvement Plans.: PIPs are discussed in 154 posts, and Performance Review is the most-named productivity item in the catalog (1,131 posts; Figure 2). These are framed almost uniformly as a step toward dismissal, not as support for improvement. Figure 5 shows the yearly volume of surveillance and RTO discussion; RTO peaks sharply in 2023 (119 posts, against 20 in 2021).

X. The Community in Its Own Words

The highest-engagement posts in each category illustrate the quantitative patterns. On the corruption of Agile estimation into surveillance:

> *"I hate Scrum. . . . Scrum seems to take away all the joy of being an engineer. Working on tasks decided by someone else, under a cadence that never stops. Counting story points and 'velocity' to measure productivity. . . "* *(r/cscareerquestions)*

On the chronic failure of OKR implementations:

> *"In 15 years, no place I've worked has ever effectively implemented OKRs. . . . Almost every place I've worked has at least mentioned, and many gave an attempt at [implementing them, but failed]."* *(r/ExperiencedDevs)*

On Goodhart's Law, named in practice if not in theory:

> *". . . the moment something becomes a metric, then it's no longer useful as a metric, because it will be gamed. Number of commits, people just start inflating their commit count. . . "* *(r/ExperiencedDevs)*

Finally, the intersection of the two themes is the most engaging content in the corpus: mental health posts that also mention metrics attract about 47% more comments than those that do not (38.8 versus 26.5 mean comments). Posts that link a developer's distress directly to being measured draw the most discussion. This connection between the two themes is exactly what the survey examines directly.

XI. Observational Answers to RQ1–RQ3

The Reddit corpus answers the three observational questions directly. **RQ1** (how frequently metrics are discussed): well-being dominates, with mental-health posts outnumbering metrics posts roughly two to one (66% vs. 34%), and the vocabulary of "productivity" itself is led by people-judging practices (performance reviews, PIPs) and individual-output metrics (lines of code, story points) rather than by systemic measures (Section V). **RQ2** (the sentiment metrics attract): framing is associated with discourse sentiment, with systemic, team-level metrics (the DORA set) drawing significantly less negativity than individual-tracking ones, and the ordering holding across categories, individual metrics, posts, comments, and two independent sentiment models (Section VI); community focus modulates the tone but does not reverse it (Section VII). The direction of this association matches what large companies report from their own internal studies: no single individual measure captures productivity [11], and developers' productivity tracks non-technical factors such as autonomy and support better than output counts [12]. **RQ3** (how the discussion evolves): conversation about measurement is growing far faster than conversation about its human cost ($+68.5\%$ vs. $+22.5\%$ over 2021–2024), and the monthly series tracks industry events such as the late-2022 layoffs and the 2023 RTO peak (Section VIII).

RQ4 and RQ5 ask what developers actually *use* and what they wish they had. The corpus contextualizes them (the adopted vocabulary clusters around evaluation practices and individual output, and well-being, though the dominant concern, has no named metric, only practices such as 1:1

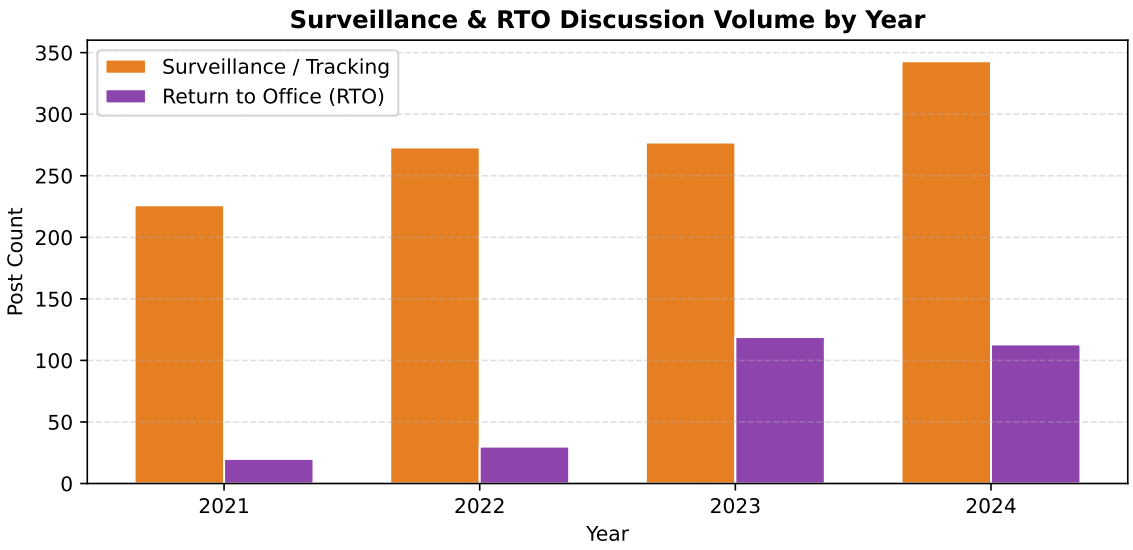


Fig. 5. Surveillance/tracking and RTO discussion volume by year (posts).

cadences), but answering them directly requires the solicited responses of the survey, which we report next (Section XII).

XII. Survey Results

The survey leg directly answers RQ4 (what developers use and how effective they find it) and RQ5 (what they wish they could measure). At the time of writing it has gathered $N = 24$ valid responses, a small convenience sample collected by snowball sampling. We treat it as descriptive evidence that triangulates the corpus and draw conclusions only where the two agree; its limits are discussed in Chapter XIII. The respondents are mostly software developers (16 of 24), with the rest split between data/AI engineers and business analysts. They lean early-to-mid career (17 of 24 have up to five years in their main role), are evenly spread across on-site, hybrid, and remote work, and come mostly from product companies (11 of 24) of varied size.

A. What Developers Use and Trust (RQ4)

a) Productivity metrics.: The adopted metrics (Figure 6, left) reproduce the corpus pattern. Story points and sprint velocity lead (10 of 24), followed by individual performance review and OKRs and by throughput (7 each); the systemic, DORA-family measures are almost absent, with change failure / MTTR named once and deployment frequency by no one. The second most common answer is *no formal metric at all* (8 of 24), which matches the corpus: productivity has no settled, named measure, just as well-being has none. When respondents rate how well the metric their team uses represents *real* productivity (Table IX), the cruder the count, the less it is trusted. Lines of code and commit counts score lowest (mean 2.5), well below story points (3.8), throughput (3.9) and code coverage (4.0). Individual OKRs and performance review are rated highest (4.4) by the developers whose teams use them. This deserves caution: an adopted metric tends to attract some after-the-fact endorsement, and the survey captures *perceived* effectiveness, not validated effectiveness. The pattern itself, though, matches prior work on how developers conceive productivity: their own definitions favor holistic, context-aware assessment over raw activity counts [13], and perceived productivity is coupled to job satisfaction more than to output volume [19].

b) Mental-health tracking.: Well-being is tracked far more loosely than output. Only 4 of 24 respondents report any *structured* tracking (surveys, formal programs); 10 describe purely informal attention (1:1s, manager conversations) and another 10 report either no tracking or never having discussed it. The single most common answer to which practices are in place is *no formal practice* (11 of 24; Figure 6, right). Where practices do exist, the human ones are trusted and the instrumented ones are not: wellbeing-focused 1:1s (mean 4.4) and therapy benefits (4.3) are rated effective, while automated pulse and climate surveys score only 2.7, barely above the 2.5 that respondents give to having no practice at all. The trusted practices are the conversational ones; the automated instruments are rated no better than nothing.

B. What Developers Wish They Could Measure (RQ5)

The open-ended answers, read thematically [31], expose a consistent gap between the metrics developers are *given* and the ones they would *choose*. Asked to name, unprompted, metrics they consider good, most respondents reach for *qualitative* dimensions: the quality and maintainability of the code, collaboration and helping peers, proactivity, and the cognitive complexity of the work. The output counts their companies adopt are largely absent from these answers, and lines of code and commits almost never appear as something a developer values. Asked what is missing, they converge on measures

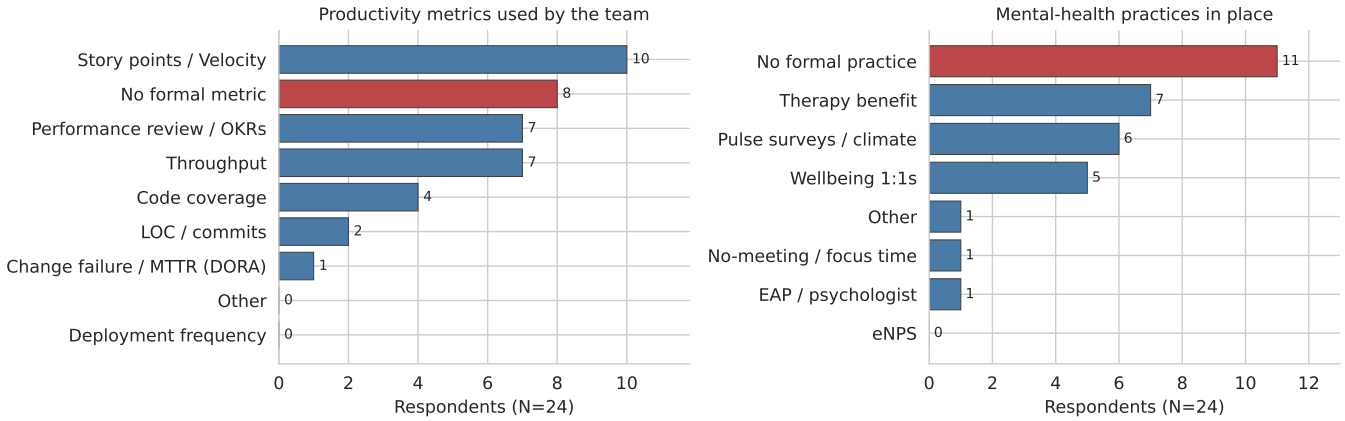


Fig. 6. Adoption of productivity metrics (left) and mental-health practices (right) among the $N = 24$ survey respondents (multiple selection allowed). The most common answer in each domain is the *absence* of a formal measure or practice (highlighted).

TABLE IX

PERCEIVED EFFECTIVENESS OF EACH METRIC/PRACTICE ON A 1–5 SCALE, RATED ONLY BY RESPONDENTS WHOSE TEAM USES IT (HENCE THE SMALL, UNEVEN n). MEANS OVER $n \leq 3$ ARE ILLUSTRATIVE, AND A VALUE WITH $n = 1$ IS A SINGLE RATING, NOT AN AVERAGE. FOR PRODUCTIVITY, 1 = "DOES NOT REPRESENT REAL PRODUCTIVITY" AND 5 = "REPRESENTS IT VERY WELL"; FOR MENTAL HEALTH, 1 = "NOT EFFECTIVE" AND 5 = "VERY EFFECTIVE".

Productivity metric	n	**Mean**	**MH practice**	n	**Mean**
Performance review / OKRs	7	4.4	Wellbeing 1:1s	5	4.4
Code coverage	4	4.0	Therapy benefit	7	4.3
Throughput	7	3.9	EAP / psychologist	1	(4.0)
Story points / velocity	10	3.8	No-meeting / focus	1	(4.0)
LOC / commits	2	2.5	Pulse / climate survey	6	2.7
Change failure / MTTR (DORA)	1	(4.0)	No formal practice	11	2.5

that weight *complexity and cognitive effort*, on rework rate, and on time-to-value over raw activity. One respondent wants "a metric that captures the time between a developer picking up a task and the feature reaching production and generating value for users"; another asks for "recovery time after a context switch", meaning how long a developer takes to be productive again after an interruption.

Several respondents go further and connect productivity measurement to well-being directly. One frames the missing metric entirely in those terms:

> Each person produces, say, X tasks per sprint on average, but no one treats moments that deviate from *her* pattern as worth worrying about. If she is delivering less, is it an interpersonal impediment with a colleague? Is she sleeping worse? Is she producing more, momentarily, under a superior's pressure? How is that affecting her mental health?[1]

The same instinct appears when respondents describe how they would *detect* a struggling colleague. Their signals of compromised mental health are read almost entirely off the work itself: a drop in the quality of deliveries and a rise in errors and rework, social withdrawal and reduced participation in meetings, irritability and impatience, visible exhaustion and stress, and overwork patterns such as "commits at dawn becoming routine, being online all the time producing little, paralysis to start tasks." The signals are symptoms in the work, and the metrics these respondents ask for are ones that would account for the person behind the numbers.

[1]Translated from the original Portuguese; see the survey instrument in Section III-C.

C. Productivity and Mental Health Are Seen as Inseparable

The survey's closing item asked, on a 1–5 scale, how related productivity and mental health are in a developer's daily work. The answer is nearly unanimous (Figure 7): the mean is 4.71, every one of the 24 respondents answered 4 or 5, 17 chose the maximum ("inseparable"), and *no one* rated the two as independent. This is the most direct support for the study's central premise that the data provide: developers do not experience productivity and well-being as separate accounts. It also lines up with the corpus, where individual-tracking metrics drew significantly more negative discourse than systemic ones. The two sources meet from opposite directions: the same practitioners who call measurement and well-being inseparable also distrust the crudest individual-output metrics and ask for measures that respect the person being measured.

XIII. THREATS TO VALIDITY

We discuss the threats to the validity of this study following the standard classification of Wohlin et al. [32].

a) Construct validity.: Sentiment, as measured by VADER, is a proxy for the construct of interest (the rejection of a metric or the distress around it), not a direct measurement

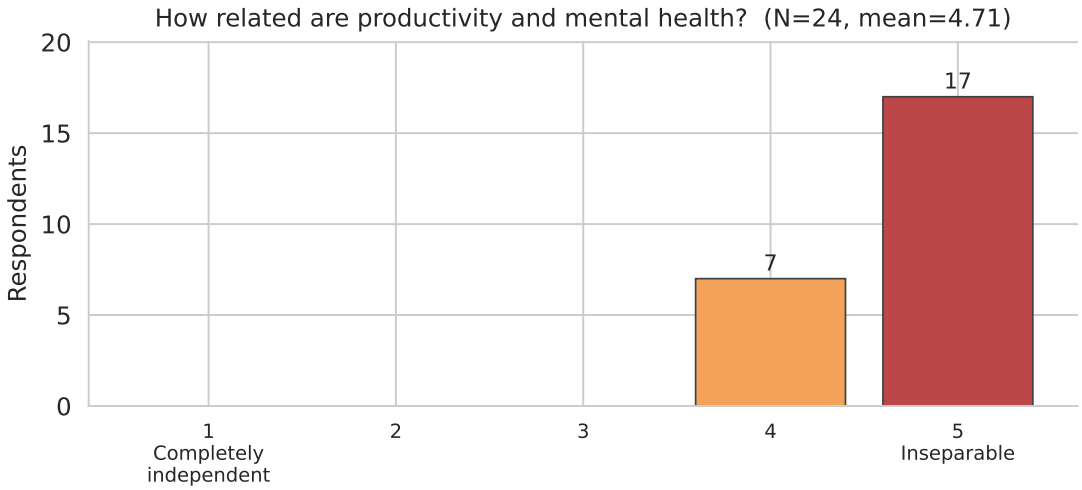


Fig. 7. Responses to "how related are productivity and mental health in a developer's day-to-day?" on a 1 (completely independent) to 5 (inseparable) scale. All 24 respondents answered 4 or 5.

of it. VADER is validated for social-media text [25], but it can misread domain-specific language, sarcasm, and technical negation; a post stating a bug was "killed" is not negative in the intended sense. We could test part of this threat: every framing-matched message was re-scored with a second model of a different family (Section VI-A), and the framing contrast holds, and even widens, under it, so the finding does not depend on VADER specifically. The two models do share one weakness, though: neither is tuned to developer vocabulary. Two preprocessing choices also bound the construct. We lower-case the text before scoring, which discards VADER's capitalization cue (it treats "TERRIBLE" as more intense than "terrible"), slightly flattening intensity; and post bodies are truncated to 500 characters, so a metric named only late in a long post can be missed. We mitigate the sentiment proxy by reporting aggregate rates over thousands of messages per category rather than individual scores, and by triangulating with the catalog and qualitative reading. The framing categories are also not mutually exclusive (Section III-A): their pairwise overlap is small (under 7%), but a doubly-matching message is counted in both, so the category sizes do not sum to the corpus. The catalog, finally, depends on an open-extraction language model, which can in principle hallucinate or miss a named item; we constrain it with a strict, exemplar-driven prompt at temperature 0, verify that it recovers all 22 items of the independent keyword baseline, and apply a manual curation pass over its output, so the published catalog is human-verified, not taken from the model unchecked.

Two further limits bound the framing–sentiment result. First, VADER scores the whole message, not the clause that names the metric, so when a person/output metric co-occurs with mental-health vocabulary (as it often does in this corpus) part of the negativity credited to the metric is really the surrounding distress. Second, the person/output category is partly defined by terms that already carry negative valence: *PIP* and *performance review* name situations that are adverse by definition, whereas the system/pipeline terms (*DORA*, *deployment frequency*) are neutral technical vocabulary. Part of the gap may therefore be lexical, a property of the words that define each category, and not of the framing they stand for; we did not correct for this, and it should temper any causal reading of the contrast. Relatedly, the system/pipeline category is operationalized almost entirely as DORA-branded vocabulary: broader pipeline notions such as cycle time and lead time, though present in the query terms, are too sparse to populate it, so "systemic" here means the DORA delivery family specifically, not every pipeline-level measure. Finally, the thematic coding of the open-ended survey answers (Section XII) was done by a single author, with no second coder or formal inter-rater agreement, so those themes are one structured reading of the responses and not independently validated categories.

b) Internal validity.: The association between metric framing and sentiment is correlational, not causal. The community result in particular is confounded: role, seniority, and demographic composition co-vary with each subreddit's topical focus, so the lower negativity in `r/devops` cannot be attributed to its metric culture alone. Including comments multiplies the sample but also pools two registers (deliberate posts and reactive replies); we therefore report the posts-only and comments-only breakdown alongside the pooled figures (Table V), and the framing ordering holds in both. The regression of Table VI further shows that the association holds after adjusting for subreddit, year, message type, and text length, though unmeasured confounds such as the author's role or seniority remain. The system/team-metric category, though robust at N=811, remains an order of magnitude smaller than the individual-metric category, reflecting its lower prevalence in the discourse.

c) External validity.: Reddit users are a self-selected, predominantly Anglophone subset of software practitioners,

and people are more likely to post about negative experiences than neutral ones, which inflates absolute negativity rates. Our claims are therefore about *relative* differences between metrics and communities, not about industry-wide prevalence. The five subreddits were chosen for role and demographic diversity but do not represent the global developer population; in particular, `r/cscareerquestions` dominates the corpus (16,446 of 25,120 posts) and skews it toward early-career and career-transition concerns. The survey is intended to counter these biases with structured, solicited responses, but at the time of writing it rests on a small convenience sample ($N = 24$), recruited by snowball sampling and skewed toward early-to-mid-career developers at product companies. It cannot, at this size, support subgroup comparisons or any claim of representativeness, and its self-selected respondents may over-represent practitioners already attentive to the metrics–well-being link. We therefore treat the survey as descriptive triangulation rather than independent confirmation, reporting per-item Likert means together with their small cell counts and foregrounding only those results that agree with the corpus. Enlarging and diversifying the sample is the primary item of future work for this leg of the study.

d) Conclusion validity.: One limitation bounds the conclusions: the archive ends in December 2024, so the window cannot be extended forward without a new data source. The corpus is also threaded, and a comment is not independent of the post it answers, so the unadjusted chi-square and Kruskal–Wallis tests ($\chi^2 = 29.7$, $\mathrm{df} = 2$; $\chi^2 = 28.4$, $\mathrm{df} = 1$; $H = 68.0$; all $p < 0.001$) overstate the effective sample size and are reported as descriptive statistics only. The primary inference comes from the logistic regression of Section VI, which clusters standard errors by thread [28] and confirms the association after covariate adjustment (OR $= 0.76$, $p = 0.003$), and from the posts-only contrast, which involves no nesting at all ($\chi^2 = 5.9$, $p = 0.015$). Given the very large samples, significance alone remains a weak guide; the effect sizes are small (Cramér's V $= 0.03$), and what supports the conclusion is the stability of the negativity gap across posts, comments, communities, covariate adjustment, and two sentiment models.

XIV. Conclusion

XV. Concluding Remarks

The analysis of the corpus answers the study's three observational questions, which the preliminary survey ($N = 24$) then complements. We summarize the conclusions by research question.

a) RQ1 (how frequently metrics are discussed).: Well-being dominates the discourse, discussed about twice as often as measurement (66% vs. 34% of posts), and the working vocabulary of "productivity" is led by evaluation practices (performance review) and individual output rather than by systemic measures.

b) RQ2 (sentiment toward metrics).: Metric framing is associated with discourse sentiment: systemic, team-level metrics (the DORA set) attract 21.9% negative discourse, against 30.8% for individual-tracking metrics and 30.9% for Scrum/Agile process, with Velocity the single most rejected metric (31.1%). The gap is statistically significant and robust: it survives a logistic regression that adjusts for subreddit, year, message type, and text length, with standard errors clustered by discussion thread (OR $= 0.76$, $p = 0.003$), and it holds across category-level and per-metric views, in posts and comments separately, and under a second sentiment model, where the gap is in fact larger. Community focus shifts the overall tone but not this ordering; we situate the finding and do not generalize it, since topic, role, and demographics are confounded across communities.

c) RQ3 (how the discussion has evolved).: Over 2021–2024 the conversation about measurement grew far faster than the conversation about its human cost ($+68.5\%$ vs. $+22.5\%$), and the monthly series bears the signature of industry events: rising volume from the late-2022 layoffs and a sharp peak in return-to-office discussion in 2023.

d) RQ4 (metrics used and their effectiveness).: The survey responses gathered so far ($N = 24$) confirm the picture the corpus suggests. Story points and sprint velocity are the most common adopted metric, individual OKRs and performance review are close behind, and the systemic DORA measures are almost never used; the second most common answer is that the team uses no formal metric at all. The crudest individual counts are also the least trusted, with lines of code and commit counts rated lowest for representing real productivity. The corpus shows the same: the adopted vocabulary clusters around people-judging practices (performance reviews, PIPs) and individual output (lines of code, pull requests), with systemic measures further down.

e) RQ5 (desirable but missing metrics).: When asked what they would rather measure, survey respondents move away from output counts and toward qualitative dimensions: code quality and maintainability, collaboration, proactivity, and measures that weight the complexity and cognitive effort of the work, rework rate, and time-to-value. Several tie productivity measurement directly to well-being, and the signals they use to spot a struggling colleague are read off the work itself (falling quality, more errors, withdrawal, overwork). In the corpus the same desire surfaces only as rejection of existing metrics and as the absence of any named well-being metric, which the survey now fills in with concrete suggestions.

Together these results substantiate the study's premise: when the industry debates productivity, it overwhelmingly measures the individual, and that individual framing is associated with more negative discourse. Well-being, though the dominant concern in these communities, is expressed almost entirely through informal narrative; it has no established metric, only practices. The gap between what is measured (individual output) and what developers talk about (sustainable, psychologically safe work) is the central tension this work documents.

f) Survey integration.: The survey closes on the study's central question. Asked how related productivity and mental health are in their daily work, all 24 respondents answered 4 or 5 on a 1–5 scale (mean 4.71), and none called the

two independent. The corpus shows that individual-tracking metrics attract more negative discourse than systemic ones, and the survey shows that the practitioners who live with those metrics distrust the crudest of them and see measurement and well-being as related. The survey is consistent with the corpus and illustrates the same pattern. Given its small, self-selected sample, it corroborates the observational findings descriptively; it does not confirm them on its own.

XVI. Future Work

Beyond completing the survey analysis, three directions follow. First, an *integrated measurement framework* that pairs systemic delivery health (DORA) with team-level well-being signals, treating the two as a single dashboard rather than separate concerns, consistent with the well-being dimension already named, but underspecified, in SPACE [9] and DevEx [10]. Second, a *longitudinal study* of teams transitioning from individual metrics (velocity) to systemic ones (DORA), measuring effects on retention and stress-related leave. Third, validation of text-based burnout signals [18] against the indicators practitioners themselves identify in survey question 13, toward non-intrusive, developer-endorsed well-being measurement.

The broader implication is that productivity measurement and developer well-being cannot remain separate management concerns. Metrics that count the individual while ignoring the conditions of their work miss the human factor, and may end up degrading the productivity they set out to measure.

XVII. Artifact Availability

Our supplementary material[2] provides the anonymized working dataset we derived (author identifiers hashed, permalinks removed), the Reddit mining and analysis scripts, and the survey instrument. The raw inputs are not redistributed: they are the public Pushshift archives [24], available independently via Academic Torrents.

References

[1] M. Strathern, "'improving ratings': Audit in the British university system," *European Review*, vol. 5, no. 3, pp. 305–321, 1997.

[2] World Health Organization, "Burn-out an "occupational phenomenon": International classification of diseases," https://www.who.int/news/item/28-05-2019-burn-out-an-occupational-phenomenon-international-classification-of-diseases, 2019, accessed: 2026-06.

[3] C. Maslach, W. B. Schaufeli, and M. P. Leiter, "Job burnout," *Annual Review of Psychology*, vol. 52, pp. 397–422, 2001.

[4] D. Graziotin, F. Fagerholm, X. Wang, and P. Abrahamsson, "What happens when software developers are (un)happy," *Journal of Systems and Software*, vol. 140, pp. 32–47, 2018.

[5] D. Graziotin, X. Wang, and P. Abrahamsson, "Do feelings matter? on the correlation of affects and the self-assessed productivity in software engineering," *Journal of Software: Evolution and Process*, vol. 27, no. 7, pp. 467–487, 2015.

[6] N. Forsgren, J. Humble, and G. Kim, *Accelerate: The Science of Lean Software and DevOps*. Portland, OR: IT Revolution Press, 2018.

[7] T. DeMarco and T. Lister, *Peopleware: Productive Projects and Teams*. New York: Dorset House Publishing, 1987.

[2]https://github.com/aisepucrio/INF1951-PROJETO-FINAL-Thomaz

[8] K. P. Pham and M. Neumann, "How to measure performance in agile software development? a mixed-method study," in *2024 50th Euromicro Conference on Software Engineering and Advanced Applications (SEAA)*, 2024, pp. 443–450.

[9] N. Forsgren, M.-A. Storey, C. Maddila, T. Zimmermann, B. Houck, and J. Butler, "The SPACE of developer productivity: There's more to it than you think," *ACM Queue*, vol. 19, no. 1, pp. 20–48, 2021.

[10] A. Noda, M.-A. Storey, N. Forsgren, and M. Greiler, "DevEx: What actually drives productivity," *Communications of the ACM*, vol. 66, no. 6, pp. 44–49, 2023.

[11] C. Jaspan and C. Sadowski, "No single metric captures productivity," in *Rethinking Productivity in Software Engineering*, C. Sadowski and T. Zimmermann, Eds. Berkeley, CA: Apress, 2019, pp. 13–20.

[12] E. Murphy-Hill, C. Jaspan, C. Sadowski, D. Shepherd, M. Phillips, C. Winter, A. Knight, E. Smith, and M. Jorde, "What predicts software developers' productivity?" *IEEE Transactions on Software Engineering*, vol. 47, no. 3, pp. 582–594, 2021.

[13] A. N. Meyer, T. Fritz, G. C. Murphy, and T. Zimmermann, "Software developers' perceptions of productivity," in *Proceedings of the 22nd ACM SIGSOFT International Symposium on Foundations of Software Engineering (FSE)*. New York: ACM, 2014, pp. 19–29.

[14] J. Azevedo, T. Sousa, J. Arriel, P. Mann, A. Garcia, and J. A. Pereira, ""i am always tired": Perceptions on mental health and productivity metrics," in *Congresso Brasileiro de Software: Teoria e Prática (CBSoft)*. SBC, 2024, pp. 79–88.

[15] J. Arriel, T. Canuto, J. Azevedo, M. V. Lima, P. Mann, A. Garcia, and J. A. Pereira, "Understanding developer well-being: measuring mental health and productivity in software teams," *Empirical Software Engineering*, vol. 31, no. 6, p. 153, 2026.

[16] N. P. Leme, J. Arriel, A. Garcia, J. Azevedo, T. Sousa, L. Bueno, J. Godinho, and J. A. Pereira, "Mental health and productivity in software development: A study with the bravo central platform," in *Proceedings of the 20th Brazilian symposium on information systems*, 2024, pp. 1–11.

[17] M. Ortu, B. Adams, G. Destefanis, P. Tourani, M. Marchesi, and R. Tonelli, "Are bullies more productive? empirical study of affectiveness vs. issue fixing time," in *Proceedings of the 12th Working Conference on Mining Software Repositories (MSR)*. Piscataway, NJ: IEEE, 2015, pp. 303–313.

[18] M. Mäntylä, B. Adams, G. Destefanis, D. Graziotin, and M. Ortu, "Mining valence, arousal, and dominance: Possibilities for detecting burnout and productivity?" in *Proceedings of the 13th International Conference on Mining Software Repositories (MSR)*. New York: ACM, 2016, pp. 247–258.

[19] M.-A. Storey, T. Zimmermann, C. Bird, J. Czerwonka, B. Murphy, and E. Kalliamvakou, "Towards a theory of software developer job satisfaction and perceived productivity," *IEEE Transactions on Software Engineering*, vol. 47, no. 10, pp. 2125–2142, 2021.

[20] R. Jongeling, P. Sarkar, S. Datta, and A. Serebrenik, "On negative results when using sentiment analysis tools for software engineering research," *Empirical Software Engineering*, vol. 22, no. 5, pp. 2543–2584, 2017.

[21] B. Lin, F. Zampetti, G. Bavota, M. Di Penta, M. Lanza, and R. Oliveto, "Sentiment analysis for software engineering: How far can we go?" in *Proceedings of the 40th International Conference on Software Engineering (ICSE)*. New York: ACM, 2018, pp. 94–104.

[22] N. Novielli, D. Girardi, and F. Lanubile, "A benchmark study on sentiment analysis for software engineering research," in *Proceedings of the 15th International Conference on Mining Software Repositories (MSR)*. New York: ACM, 2018, pp. 364–375.

[23] A. Edmondson, "Psychological safety and learning behavior in work teams," *Administrative Science Quarterly*, vol. 44, no. 2, pp. 350–383, 1999.

[24] J. Baumgartner, S. Zannettou, B. Keegan, M. Squire, and J. Blackburn, "The Pushshift Reddit dataset," in *Proceedings of the 14th International AAAI Conference on Web and Social Media (ICWSM)*. Palo Alto, CA: AAAI, 2020, pp. 830–839.

[25] C. J. Hutto and E. Gilbert, "VADER: A parsimonious rule-based model for sentiment analysis of social media text," in *Proceedings of the 8th International AAAI Conference on Weblogs and Social Media (ICWSM)*. Palo Alto, CA: AAAI, 2014, pp. 216–225.

[26] F. Barbieri, J. Camacho-Collados, L. Espinosa Anke, and L. Neves, "TweetEval: Unified benchmark and comparative evaluation for tweet classification," in *Findings of the Association for Computational Linguis-*

tics: EMNLP 2020. Stroudsburg, PA: Association for Computational Linguistics, 2020, pp. 1644–1650.

[27] J. Cohen, "A coefficient of agreement for nominal scales," *Educational and Psychological Measurement*, vol. 20, no. 1, pp. 37–46, 1960.

[28] A. C. Cameron and D. L. Miller, "A practitioner's guide to cluster-robust inference," *Journal of Human Resources*, vol. 50, no. 2, pp. 317–372, 2015.

[29] N. E. Gold and J. Krinke, "Ethics in the mining of software repositories," *Empirical Software Engineering*, vol. 27, no. 1, p. 17, 2022.

[30] B. A. Kitchenham and S. L. Pfleeger, "Personal opinion surveys," in *Guide to Advanced Empirical Software Engineering*. London: Springer, 2008, pp. 63–92.

[31] V. Braun and V. Clarke, "Using thematic analysis in psychology," *Qualitative Research in Psychology*, vol. 3, no. 2, pp. 77–101, 2006.

[32] C. Wohlin, P. Runeson, M. Höst, M. C. Ohlsson, B. Regnell, and A. Wesslén, *Experimentation in Software Engineering*. Berlin: Springer, 2012.